



Computer Networks Lab

CL-3001 | Fall 2026 | Section BAI-5A

Course Instructor: Mr. Abdullah Shaikh

LAB # 05 — TASKS

Introduction to FTP, SMTP and their Implementation

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Section	BAI-5A	Date of Submission	

Submission Guidelines

- Complete all tasks and paste the screenshots of every configuration and every test into this Word file.
- Submit the completed file as a **PDF** on GCR, named **LabCourseName_Lab#-YourName YourRollNo/ID** — e.g. **CN_Lab#05-Abdullah Shaikh-25K-7826**.
- Upload all **.pkt** and **.pcap** files to your Google Drive and grant access to abdullah.shaikh@nu.edu.pk.
- The router in every topology must be configured through the CLI, not the GUI.

Label every device on the workspace with its IP and MAC address using **Place**

Note. • *No deadline will be extended under any circumstances.*

OBJECTIVE

To understand how the File Transfer Protocol (FTP) and the Simple Mail Transfer Protocol (SMTP) work at the application layer, to implement FTP, and SMTP services on a routed network in Cisco Packet Tracer.

TASK # 01 — FTP SERVER ON A ROUTED NETWORK

Design and implement the topology given in Figure 1 in Cisco Packet Tracer. The network consists of two subnets joined by a router: a client LAN and a server LAN that hosts the FTP server. The router must be configured through the CLI. Attach all the necessary screenshots in this Word file and upload the PKT file to your drive.

Addressing Requirement

Derive both network addresses from the last four digits of your roll number:

- Client LAN (Fa0/0 side): **XX.XX.0.0 / 16** — e.g. roll number 4129 → **41.29.0.0 / 16**

- Server LAN (Fa0/1 side): **XX.0.0.0 / 8** — e.g. roll number 4129 → **29.0.0.0 / 8**

Every host must be given a static IP address, the correct subnet mask and the correct default gateway.
Complete Table 1 before you begin configuring the devices.

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Fa0/0	25.39.0.1	255.255.0.0	—
Router0	Fa0/1	39.0.0.1	255.0.0.0	—
PC0	Fa0	25.39.0.10	255.255.0.0	25.39.0.1
PC1	Fa0	25.39.0.11	255.255.0.0	25.39.0.1
PC2	Fa0	25.39.0.12	255.255.0.0	25.39.0.1
PC3	Fa0	39.0.0.10	255.0.0.0	39.0.0.1
FTP Server	Fa0	39.0.0.2	255.0.0.0	39.0.0.1

Table 1: IP addressing table for Task # 01

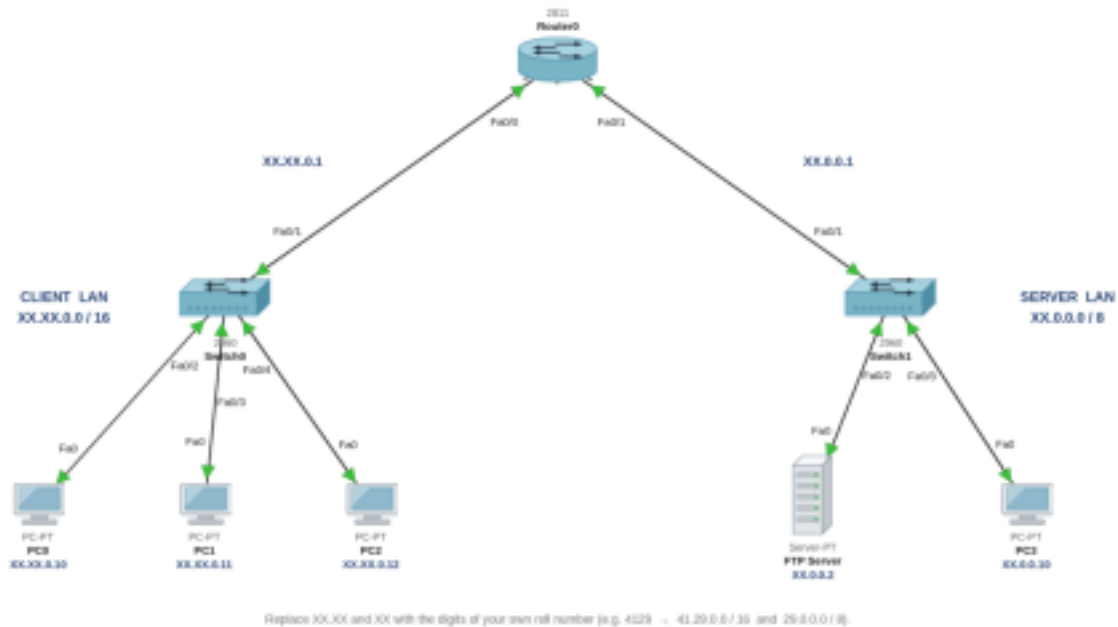


Figure 1: Topology for Task # 01

Implementation Steps

1. Build the topology shown in Figure 1 using one 2811 router, two 2960 switches, four PCs and one Server-PT. Connect all devices with straight-through copper cables.

2. Configure both router interfaces through the CLI (replace the placeholders with your own addresses):
3. Assign the static addresses from Table 1 to every PC and to the server (Desktop → IP Configuration). Verify inter-subnet connectivity by pinging the FTP server from PC0 and from PC2.

4. On the server, open **Services** → **FTP**. Turn the service **On**, then create a user account in which the user name is your first name and the password is your roll number. Grant the Write, Read, Delete, Rename and List permissions, click **Add**, and remove the default **cisco** account.
5. On PC0 open **Desktop** → **Text Editor**, type your name, roll number and section, and save the file as **<YourRollNo>.txt**.
6. Upload the file from PC0 using the Command Prompt
7. Download the same file from a PC on the other subnet (PC3), then delete one file from the server and list the directory again:
8. Label every device on the workspace with its IP and MAC address using **Place Note**, and save the file as **CN_Lab05_Task01_<YourRollNo>.pkt**.

Screenshots Required

- Output of show ip interface brief and the running configuration of the router.
- Successful ping from a client on the client LAN to the FTP server.
- The FTP service page showing your user account and its permissions.
- The complete FTP session on both subnets, showing the login message, dir, put, get and delete outputs.

PASTE YOUR CLI CONFIGURATION, FTP SERVER SETUP AND CLIENT TEST SCREENSHOTS HERE

 Router0

Physical Config CLI Attributes

```
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 25.39.0.1 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface fa0/1
Router(config-if)#ip address 39.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	25.39.0.1	YES	manual	up	up
FastEthernet0/1	39.0.0.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 39.0.0.2

Pinging 39.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 39.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 39.0.0.2

Pinging 39.0.0.2 with 32 bytes of data:

Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 39.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

FTP-server

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP
- PRP

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☒ Write ☒ Read ☒ Delete ☒ Rename ☒ List

	Username	Password	Permission
1	kashif	2539	RWDNL

Add Save Remove

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 39.0.0.2

Pinging 39.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127
Reply from 39.0.0.2: bytes=32 time<1ms TTL=127

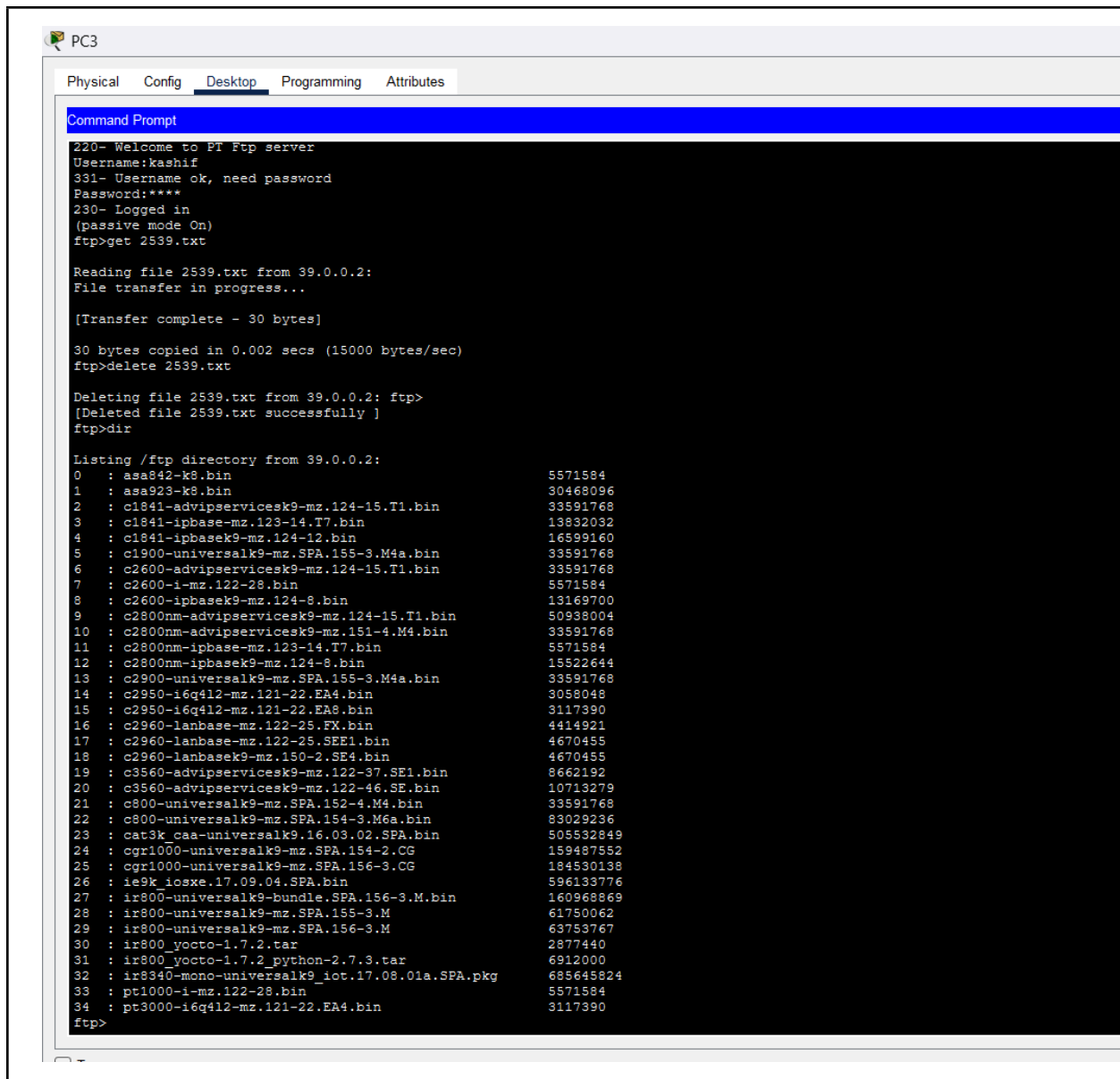
Ping statistics for 39.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ftp 39.0.0.2
Trying to connect...39.0.0.2
Connected to 39.0.0.2
220- Welcome to FT Ftp server
Username:kashif
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>put 2539.txt

Writing file 2539.txt to 39.0.0.2:
File transfer in progress...

[Transfer complete - 30 bytes]

30 bytes copied in 0.081 secs (370 bytes/sec)
ftp>
```



```
PC3
Physical Config Desktop Programming Attributes
Command Prompt
220- Welcome to PT Ftp server
Username:kashif
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>get 2539.txt

Reading file 2539.txt from 39.0.0.2:
File transfer in progress...

[Transfer complete - 30 bytes]

30 bytes copied in 0.002 secs (15000 bytes/sec)
ftp>delete 2539.txt

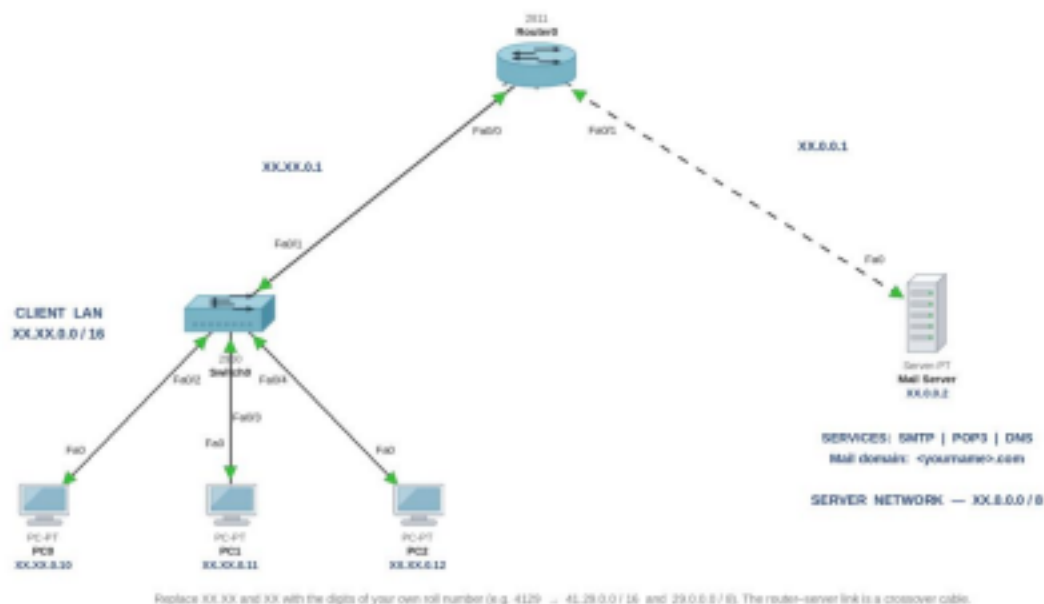
Deleting file 2539.txt from 39.0.0.2: ftp>
[Deleted file 2539.txt successfully ]
ftp>dir

Listing /ftp directory from 39.0.0.2:
0 : asa842-k8.bin 5571584
1 : asa923-k8.bin 30468096
2 : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
3 : c1841-ipbase-mz.123-14.T7.bin 13832032
4 : c1841-ipbasek9-mz.124-12.bin 16599160
5 : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
6 : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
7 : c2600-i-mz.122-28.bin 5571584
8 : c2600-ipbasek9-mz.124-8.bin 13169700
9 : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
10 : c2800nm-advipservicesk9-mz.151-4.M4a.bin 33591768
11 : c2800nm-ipbase-mz.123-14.T7.bin 5571584
12 : c2800nm-ipbasek9-mz.124-8.bin 15522644
13 : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
14 : c2950-16q412-mz.121-22.EA4.bin 3058048
15 : c2950-16q412-mz.121-22.EA8.bin 3117390
16 : c2960-lanbase-mz.122-25.FX.bin 4414921
17 : c2960-lanbase-mz.122-25.SEE1.bin 4670455
18 : c2960-lanbasek9-mz.150-2.SE4.bin 4670455
19 : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
20 : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
21 : c800-universalk9-mz.SPA.152-4.M4a.bin 33591768
22 : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
23 : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
24 : cgr1000-universalk9-mz.SPA.154-2.CG 159487552
25 : cgr1000-universalk9-mz.SPA.156-3.CG 184530138
26 : ie9k_iosxe.17.09.04.SPA.bin 596133776
27 : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
28 : ir800-universalk9-mz.SPA.155-3.M 61750062
29 : ir800-universalk9-mz.SPA.156-3.M 63753767
30 : ir800_yocto-1.7.2.tar 2877440
31 : ir800_yocto-1.7.2_python-2.7.3.tar 6912000
32 : ir8340-mono-universalk9_iot.17.08.01a.SPA.pkg 685645824
33 : pt1000-i-mz.122-28.bin 5571584
34 : pt3000-16q412-mz.121-22.EA4.bin 3117390
ftp>
```

TASK # 02 — SMTP AND POP3 EMAIL SERVICE

Design and implement the topology given in Figure 2. The mail server is connected directly to the router with a copper cross-over cable and provides three services: SMTP, POP3 and DNS. The clients must send and receive email using a domain name, not an IP address. The router must again be configured through the CLI.

Use the same roll-number-based addressing scheme as Task # 01: the client LAN is XX.XX.0.0 / 16 and the server network is XX.0.0.0 / 8.



7. From PC0, compose and send an email to PC1 with the subject **CL-3001 Lab 05 – <Your Name> – <Your Roll No>**. On PC1 click Receive, open the message and reply to it. Confirm that PC0 receives the reply.
8. Send a second email from PC1 to PC2 to prove that all three accounts work, then attempt to send an email to a user that does not exist on the server and record the error message that the client reports.
9. Label all devices with their IP and MAC addresses using **Place Note** and save the file as **CN Lab05 Task02 <YourRollNo>.pkt**.

Email Account Plan

Complete Table 2 with the accounts you created on the mail server.

Device	Display Name	Email Address	Incoming / Outgoing Server	Password
PC0	PC0	PC0@kashif.com	mail.kashif.com	123
PC1	PC1	PC1@kashif.com	mail.kashif.com	456
PC2	PC2	PC2@kashif.com	mail.kashif.com	789

Table 2: Email accounts configured for Task # 02

Screenshots Required

- The EMAIL service page showing the domain name and the list of user accounts.
- The DNS service page showing the A record for the mail server.
- The mail client configuration of at least two PCs.
- The sent message, the received message on the destination PC, the reply, and the error message produced for the unknown recipient.

PASTE YOUR SERVER CONFIGURATION, MAIL CLIENT SETUP AND EMAIL TEST SCREENSHOTS HERE

```
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
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249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

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Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

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Router(config-if)#exit
Router(config)#interface fa0/1
Router(config-if)#ip address 39.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 25.39.0.1       YES manual up          up
FastEthernet0/1 39.0.0.1        YES manual up          up
Vlan1          unassigned      YES unset  administratively down down
Router#
```

Compose

Reply

Receive

Delete

Configure Mail

	From	Subject	Received
1	PC1@kashf.com	RE: CL-3001 Lab 05-Kashf-2539	Wed Oct 7 2026 00:47:25

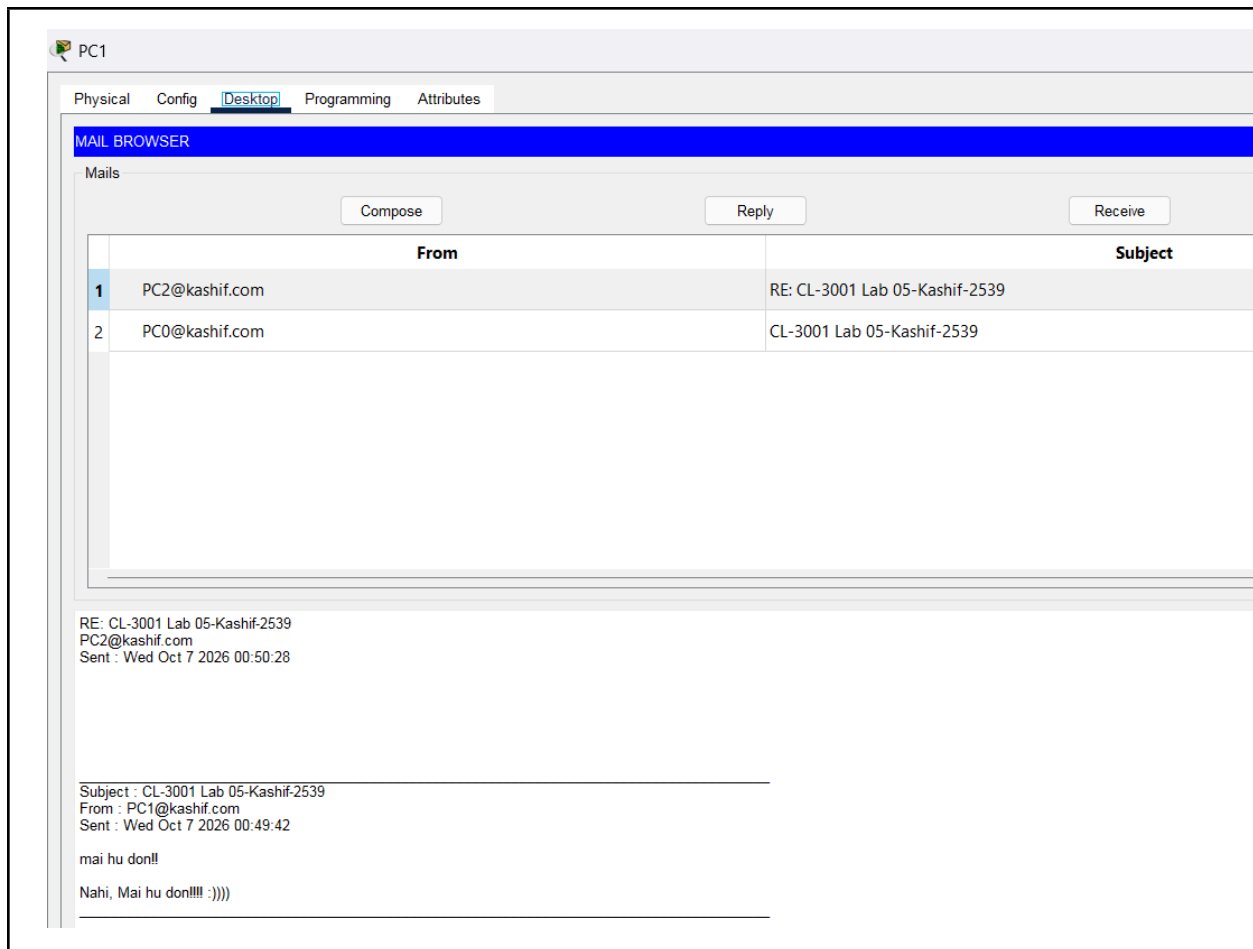
RE: CL-3001 Lab 05-Kashf-2539
PC1@kashf.com
Sent: Wed Oct 7 2026 00:47:25

Subject: CL-3001 Lab 05-Kashf-2539
From: PC0@kashf.com
Sent: Wed Oct 7 2026 00:46:45

Hello jael
wassup DUDE?

Receiving mail from POP3 Server mail.kashf.com
DNS resolving. Resolving name: mail.kashf.com by querying to DNS Server: 39.0.0.2 DNS resolved ip address: 39.0.0.2
Receive Mail Success.

Cancel
Send/Receive



The End!

Good luck — remember to upload your PKT files along with this document.